

Acknowledgements



*Nei rā te kaupapa o te reo māori e / He taonga tuku iho nō nga tipuna
he parekarakawa, he mea hirahira e / Toku reo, toku ohoohe mapihi maurea whakakaimarihi*

— NEI RĀ TE KAUPAPA, BY TE OKEROA HURIWAKA AND TAWHIRO MAXWELL

Waipapa Taumata Rau (The University of Auckland) sits within the rohe of several Māori iwi/hapū, including Ngāti Whātua, Ngāti Whātua o Ōrākei, Te Kawerau a Maki, Ngāti Tamaoho, Te Ākitai Waiohū, Ngāti Maru, Te Patukirikiri, Ngāti Pāoa, Ngāti Tamaterā, Ngāi Tai ki Tāmaki, Ngāti Te Ata, and Ngāti Whanaunga, and Waikato-Tainui. Ipipiri (the Bay of Islands) sits within the rohe of Ngāpuhi and Ngāti Rēhia. I acknowledge these iwi and hapū as the tāngata whenua of Tāmaki Makaurau, Ipipiri, and Aotearoa, I acknowledge that sovereignty was never ceded under Te Tiriti O Waitangi, and I pay my respects to kuia and kaumātua of these iwi and hapū, past, present, and future. Tūturu whakamaui, kia tina! Tihei mauri ora!

And I came from the Dreaming / From the dusty red-soil plain

I am the ancient heart / The keeper of the flame.

*I stood upon that rocky shore / I watched those tall ships come
when for eighty thousand years I've been / The first Australian*

— I AM AUSTRALIAN, AS PERFORMED BY THE BUSHWACKERS

Throughout my travels for this PhD, I was privileged to visit several lands I never dreamed I could as a child. In Australia, I worked on lands traditionally occupied by peoples of the Kulin Nations (predominantly the Wurundjeri people), and the Jagera, Yuggera and Ugarapul peoples. I acknowledge their sovereignty and traditional ownership and custodianship of the lands and waterways on which Swinburne University of Technology and the University of Southern Queensland (Springfield) are located. I acknowledge their Elders, past, present, future and emerging, and I acknowledge that sovereignty was never ceded. I thank the kind hospitality of the Polish, Belgian, Singaporean, German, Dutch, and Australian peoples, all of whose lands I visited throughout this thesis. I have learned so much, sitting on the shores of the Eternal Sea that is the

Milky Way.

The author wishes to acknowledge the use of New Zealand eScience Infrastructure (NeSI) high performance computing facilities, consulting support and/or training services as part of this research. New Zealand’s national facilities are provided by NeSI and funded jointly by NeSI’s collaborator institutions and through the Ministry of Business, Innovation & Employment’s Research Infrastructure programme. URL <https://www.nesi.org.nz>.

This work made use of v2.2 of the Binary Population and Spectral Synthesis (BPASS) models as described in [Eldridge et al. \(2017\)](#) and [Stanway & Eldridge \(2018\)](#).

*Raise a glass to all of us
Tomorrow there’ll be more of us
Telling the story of tonight.*

— FOUND/TONIGHT (BEN PLATT AND LIN-MANUEL MIRANDA)

I am the proud grandchild of immigrants, and by my count, the first person who identifies as Sāmoan to graduate with a PhD from the Department of Physics at the University of Auckland in (at least) a quarter century. I’m also one of the few gender non-conforming students in my Department. To all my fellow BIPOC students, all my queer whānau, and all my other underrepresented students in the sciences, I see you, I recognise you, I feel you. Kia kaha, kia māia, kia manawanui.

It would be impossible for me to list my entire whakapapa. It is known as far back as nine generations in parts, tracing back to around the mid-1700s – including to the first Lord Mayor of the City of Sheffield, a Prime Minister of Samoa, several Ministers in Samoa – including a former Minister of Finance – and Chair of Samoa’s National Bank, and some of the major clans in Samoa. We suspect there may be an American President in there too. At last count we know of about 500 relatives – my Great-Great-Aunt’s father-in-law had thirteen children in the mid-1800s. But I want to especially mihi to my mother Sue, my late aunt Leslie, my late grandfather Alan, my grandmother Nanette and her late mother Mary “Sugalu” Scott, my aunt Alison and uncle Pete, my great-uncle Ben and great-aunt Lulu, my cousins Jeremy, Jaime, Charlotte, Candice, Ngareta, Wairaka, Marihi, and Tama. To the rest of my ‘aiga: *e lautele le vateatea, ae le o le avanoa iinei: Ou te alofa ia te outou uma.*

Life before death. Strength before weakness. Journey before destination.

— FIRST IDEAL OF THE KNIGHTS RADIANT

For Jaffe, Bailey, Riegel, Ray, Mercer, Johnson, Willingham, and O’Brien: Vox Machina, the Mighty Nein, Bells Hells, Gilmore, Allura, Delilah, Pumat, Xandis, and Ryn. For Iyengar, Carazo, Ray, Riegel, Willingham, Wilson, and Mulligan: the Ring of Brass, Vespene, Maya, and Kir. For Lewis: Peter, Edmund, Susan, Lucy, Aslan and Jadis. For Sanderson: Kaladin, Syl, Shallan, Jasnah, Vin, Elend, Kelsier, Rysn, and Hoid. For Vincke, Wilde, English, Newbon, Downie, Solomon, Béart, Jones, Wiles, Mercer, Gregory, Simmons, Isaacs,

and Robertson: Lae'zel, Shadowheart, Astarion, Gale, Wyll, Karlach, Halsin, Jaheira, Minsc, Minthara, Ketheric, Gortash, and Orin the Red. For those who imagine the strange and beautiful and complex and weird and mystical worlds of fantasy, whether betwixt the pages of a book, on the silver screen, or upon the roll of a d20, I pay my enormous regards to you.

I argue that we choose to be good because of our bonds with other people and our innate desire to treat them with dignity. Simply put, we are not in this alone.

— CHIDI ANAGONYE

It occurs to me that there are precious few moments in one's life to acknowledge and thank the people who contributed to where one is now. I hold the belief that a person is defined by the sum of their actions and interactions, so I would like to acknowledge a few of the people that have made me the person I am today.

I'd like at this point to acknowledge friends: Connor, Frazer, Girish, Hunter, Pierce, Ryan, Scott, and Sean, with whom I shared some of my first physics and mathematics courses. Emma, the self-appointed creative director of this thesis. Josh, Tessa, Hamish, Jasmine, Ryan: my first experience with D&D was with Eldrin. Andrew, George, Hamish, Connor, Callum, and Mark: Katerina was my longest player experience; playing her taught me so much about myself (and Andrew). Josh, Liam, Arthur, Connor, Andrew, Oliver: Ayo and the rest of the Blue Originators (you never learned their party's name) owe you a debt of gratitude. Even good old Necromancer Bob says hi. Connor, Kali, Jackson, Andrew: I can't wait for you to see what other war crimes Matilda has in store.

I'm going to thank Jasmine again. It is a well kept secret in our friend group that Jasmine got top in course in Linguistics 203 (Applied English Grammar) during undergrad. She has helped me time and time again, and especially during the writeup of the paper version of Chapter 3. Ngā mihi e hoa.

Thank you to my office mates, Max, Bushra, Jonathan, Petra, Hamish, Erin, and Laura. On the topic of Big Dog and Big L: I can't say what I got told they'd do if I didn't acknowledge them. They know what they said. But thank you for long walks on beaches, fish and chips, drinks at My Bar (we need to get a better bar), and generally being a welcome distraction whilst writing this thesis. Also, get therapy, the both of you.

Thank you to Ruby Davtyan, a visiting researcher from Lund University. I don't speak Russian, and Ruby does, so they offered to (roughly) translate the original [Petrov \(1946\)](#) manuscript when I finally got my hands on it, even though astronomy (let alone binary stars!) is not their field as a nanophotonics researcher.

Once a king or queen in Narnia, always a king or queen.

— ASLAN

I've lost touch with some of the people I did high school with but the formative experiences there still sit inscribed in indelible ink in my mind. Caitlin, who has been one of my best friends for the better part of a quarter-century (and who would *not* let me live it down were I to forget her). Callum, Susannah, Kyle, Grayson, Tomek "if you're going to try to pronounce my last name, put some **EMPHASIS** on it" Pietkiewicz

and the entire business and commerce crew. Sam, Louise, Crooks, the IT Ninjas. Isabella, Chyna, Evan, Parth, Jay, Matt, all the people at NZSDC 2015, even if I had to argue in favour of alternative medicine and exclusive schools for LGBTQIA youth, those moments will be with me forever. Finn and Susannah: I know they said they never ranked us in maths classes, but they ranked us in maths classes, and I know we always vied for the top three places. Finn is an architect now and Susannah is admitted to the High Court – does that mean I finally win the ranking?

I'd also like to thank Di Eastwood, Maddy Haydock, Melissa Johansson, and Lauren Mulcahy (our debating coach, and my Year 13, 12, and 11 English teachers). Not all of you are teaching anymore, but I still remember vivid parts of your lessons. Miss Mulcahy commending my essay structure (for not opening an essay with "The question I am answering is..." but actually motivating why I enjoyed *To Kill a Mockingbird* and *Romeo+Juliet*). Miss Johansson, for showing faith in the process of my creative writing portfolio ("Sean, I want to send you to counselling for this story. Change nothing, this is an E."). Mrs Haydock, firstly for Mufasa but also for believing in my weird and experimentalist poetry (I was going through a bit of a fourth-wall breaking phase with poetry), for letting me rant for 30 minutes on why *The Matrix* is actually a Marxist allegory and the (concerningly frequent) themes from Freudian psychoanalysis we can pull from it. Moreover, I remember in English class, you congratulating me for using "moreover" correctly in a sentence. Mrs Eastwood for always challenging me to do better in debating. I've said it before and I will say it again: I credit a mix of you four for my erudition and eloquence.

On the topic of teachers, I would like to mihi to Robbie Lumkong, Andy Dyson, Chris Whitaker. Physics lessons in Year 12 and 13 with Mr Lumkong (and in Year 13 with Mr Dyson) were absolutely batshit crazy – can I say batshit in a thesis? – but it is the undeniable infective enthusiasm for physics that someone who has played international rugby¹ brings that made me even think about doing PHYSICS 120 when I came to university and had a free paper in first year. Mr Whitaker, I still remember sitting on a bus behind you in Year 11 Physical Science and asking you a tonne of questions about the big bang and the standard model.

Right, I'm almost done here, I swear.

Physics is a collaborative endeavor, and I would be remiss to not acknowledge my research compatriots. Thank you to the BPASS team, both in Auckland and abroad: John,² Heloise, Max, Sohan, Wouter, Petra, Erin, Adib, Conor, Gareth, Ashley, and Ankur. We've also been joined by a few students who have since left the team (because they graduated!): Gleb, Emma, Louis, Ethan, Siris, Celina, and Amanda. Thank you to our PIs, Dr Elizabeth Stanway and Prof. Jan Eldridge.

Speaking of Jan! Thank you to Prof. Jan Eldridge and Assoc. Prof. Nick Rattenbury. I genuinely cannot image a more perfect supervision team. I have learned an immense amount from each of them. You'll notice that all of my contractions of people's titles either omit (in the case of "Dr") or contain periods (in the case of basically every other title). Nick was very pedantic about this in my Masters thesis and now I subconsciously check for this when reading other people's prose. Thank you by extension to Prof. Richard Easter. Jan,

¹This is not a joke or a quip, my Year 13 physics teacher played as a number eight in the 1999 Rugby World Cup for the USA. They lost 53-8 to Ireland, 27-25 to Romania, and 55-19 to Australia. They also lost 106-8 to England, outside of the RWC.

²John is responsible for me adding exactly one of the quotes in this thesis. I'll leave the reader to guess which one.

Nick, and Richard were my lecturers for PHYSICS 107/G in Semester Two of 2016. I needed extra papers to fill out my first year, as there were only three required papers for my (at that time) Computer Science degree. I took PHYSICS 120 and 150 because I was good at it in High School (see Mr Lumkong, Mr Dyson, and Mr Whitaker above), and PHYSICS 107/G because it sounded fun. I left that course a physics double major, and no joke, restructured my entire degree around taking PHYSICS 356. Jan, Nick, and Richard are a large part of why I am a scientist today. I cannot thank them enough for that. From the bottom of my heart, thank you, ngā mihi, vinaka vakalevu, fa'afetai lava.

I'd also like to mihi to a few of my inspirational heroes in science as a whole. Dr Derek Miller (Veritasium), Mrs Dianna Cowern (PhysicsGirl), Mr Destin Sandlin (SmarterEveryDay), Mr John D. Boswell (melodysheep), and Mr Tim Blais (acapellascience).

*That you are here - that life exists, and identity;
that the powerful play goes on and you may contribute a verse.*

— O ME! O LIFE! BY WALT WHITMAN

When I started this PhD, I had two alive grandparents. I now have only one, my grandmother Nanette. For Alan, Walter, and Eileen, I carry your memories with me always. In the words of Liam O'Brien: *I was richer for having known you, friend.*

I could not be more keen, for some plain old boring precedented times.

— CR. RICHARD HILLS

I've noticed that the more theses I write, the more people I thank. My PGDipSci dissertation thanked five people. My Masters thesis thanked some fifty-five, a Government, and several coffee companies. Did I beat it this time?

Run, you clever [girl], and be a Doctor.

— CLARA OSWIN OSWALD, DOCTOR WHO

*No, thank YOU. One thing about trains... it doesn't
matter where they're going. What matters is deciding
to get on.*

— THE CONDUCTOR, THE POLAR EXPRESS

